

[Skip to main content](#)

- [Welcome](#)
- [Sample Content](#)
 - [Meteo 101](#)
 - [Meteo 241](#)
 - [Meteo 361](#)
 - [Meteo 410](#)
- [Our Alumni](#)
- [Join Us!](#)

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Search form

Search

Sample Content

- [Meteo 101: Sample Content](#)
- [Meteo 241: Sample Content](#)
- [Meteo 361: Sample Content](#)
 - [Lesson 1. Meeting the Mesoscale](#)
 - [Lesson 3. Sizing up the Synoptic Scale](#)
 - [Assessing Strong Updrafts](#)
 - [Lapse Rates](#)
 - [Vertical Wind Shear](#)
 - [The Big Picture at 500 mb](#)
 - [Lee Troughs](#)
 - [Surface Boundaries and the Big Picture](#)
 - [Prefrontal Troughs and Confluence](#)
 - [The Big Picture at 850 mb](#)
 - [Elevated Convection](#)
 - [Upper-Level Jet Streaks](#)
 - [Coupled Jet Streams](#)
- [Meteo 410: Sample Content](#)

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- [login](#)

The Big Picture at 850 mb

This is a sample lesson page from the [Certificate of Achievement in Weather Forecasting](#) offered by the Penn State Department of Meteorology. Any questions about this program can be directed to: [Steve Seman](#)

Prioritize...

When you've completed this page, you should be able to define elevated thunderstorms, and describe what forecasters look for at 850-mb when assessing the big picture in making a forecast for deep, moist convection. Namely, you should be able to discuss the impacts of 850-mb warm advection and low-level jet streams.

Read...

So far, we've covered the roles of the 500-mb pattern and surface convergence in the development of deep, moist convection. Now it's time to see what forecasters look for at another level in the lower troposphere--850 mb. What aspects of the 850-mb pattern provide clues for forecasters about the development of thunderstorms?

In short, forecasters are primarily looking for two things at 850 mb--evidence of warm advection and the presence of low-level jet streams (ribbons of relatively fast winds in the lower troposphere). Why are these things important to forecasters? Let's investigate, starting with 850-mb warm advection.

You learned in your previous studies that the strongest warm advection associated with a mid-latitude low-pressure system occurs along and north of the low's warm front (not in the warm sector, which tends to be relatively homogeneous with regard to temperature). You also learned that warm advection north of a warm front goes hand in hand with overrunning. Thus, the footprint of overrunning at 850 mb is, not surprisingly, a pocket of warm advection. Below, the 14Z 850-mb analysis on July 14, 2010, shows that there was fairly strong warm advection (red shading) over Minnesota at this time.

 **The 14Z analysis of 850-mb heights, winds, temperature, and temperature advection on July 14, 2010**
The 14Z 850-mb analysis on July 14, 2010. Dark contours denote 850-mb heights. Dashed, red contours are 850-mb isotherms. Wind barbs represent 850-mb winds. Contours filled with blue indicate areas of cold air advection, while red color-filled contours designate areas of warm air advection. Note the rather strong warm advection over Minnesota at this time.

Credit: Storm Prediction Center

This pocket of warm advection was located north of a warm front ([12Z surface analysis](#)), and was closely connected to a cluster of thunderstorms that had developed (check out the [14Z regional radar mosaic](#)). Indeed, the severe weather over the Upper Mississippi Valley on July 14, 2010 that you've studied over the past few sections, was not confined to the afternoon hours. Recall that deep, moist convection is possible north of a warm front when there's [upper-level divergence above the sloping warm front](#), where warm-air advection is occurring in concert with overrunning, and that's precisely what happened on the morning of July 14, 2010. Note, however, that the updrafts for such thunderstorms do not originate at the surface. Instead, they originate above the cold stable layer at the surface, and are appropriately called **elevated thunderstorms**.

Formally, an elevated thunderstorm is a type of deep, moist convection whose updraft originates above the planetary boundary layer. In contrast, updrafts associated with surface-based convection originate at the ground. As a general rule, elevated convection develops above a stable layer of air in the lower troposphere, which means either above a nocturnal inversion, or on the cold side of an anafront (usually warm front or stationary front).

You may have been surprised to see me mention "nocturnal inversions" because you may recall that they typically form on clear nights with calm winds (not exactly the types of nights that make you think "thunderstorms"). But, keep in mind that the entire night need not be clear for nocturnal inversions to form. Indeed, nocturnal inversions can form rather quickly after sunset (a few hours), and the evolving weather pattern could then favor elevated thunderstorms later in the night.

To get a better idea about the contrast between elevated thunderstorms and surface-based thunderstorms, check out the idealized skew-T diagrams below. The sounding on the left is consistent with elevated convection. Note the stable layer (a layer of relatively warm air) between 900 mb and 750 mb (roughly). If unstable parcels of air are lifted from the top of this stable layer, they become positively buoyant, setting the stage for elevated convection (although CAPE is rather small). The sounding on the right favors surface-based convection (surface air parcels lifted to the LFC become dramatically positively buoyant through a deep layer).

 **(Left) An idealized skew-T showing an elevated convection set up. (Right) An idealized skew-T showing a surface-based convection set up.**

(Left) If unstable parcels of air are lifted from the top of the stable layer between 900 mb and 750 mb (approximately), they become positively buoyant, setting the stage for elevated convection (even though CAPE is rather small). (Right) This sounding favors surface-based convection (surface air parcels lifted to the LFC become dramatically positively buoyant through a deep layer). Note that CAPE is noticeably larger.

Credit: David Babb

So, forecasters look for pockets of warm advection at 850-mb because they can signal overrunning. If lapse rates above the cool, stable layer at the surface are sufficiently steep, elevated thunderstorms can develop, especially if some upper-level divergence is present to give parcels an additional kick of upward motion to get things started.

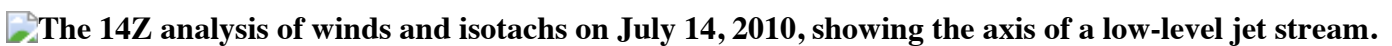
Low-Level Jet Streams and Moisture Convergence

On the morning of July 14, 2010, we can see the source of the extra kick of upward motion from upper-level divergence on the [12Z NAM model 500-mb analysis of heights and vorticity](#). Note the weak vorticity maxima present ahead of the closed low centered in south-central Canada.

So, the development of elevated convection in this case was fairly textbook, fitting the [conceptual model](#) that you learned about previously. These thunderstorms undoubtedly benefited from the presence of a **low-level jet stream** over the region, as well. Formally, low-level jet streams are ribbons of relatively fast winds in the lower troposphere driven by strong height gradients.

Why are low-level jet streams of interest? First, they can efficiently usher moist air into a region (remember that 850 mb is located in the lower troposphere, where most atmospheric water vapor is located). Moist, boundary-layer air rushing into a region can go hand-in-hand with deep moisture convergence. On July 14, 2010, we can see that the low-level jet stream was driven by strong 850-mb height gradients, as shown on the [12Z NAM model analysis of 850-mb heights, temperatures, and winds](#). From this analysis, should also be able to diagnose the warm advection you saw previously over Minnesota. Note how the 850-mb wind barbs blow across the isotherms (thin, red contours) from higher to lower values.

To better help you focus on the corridor of fast 850-mb winds marking the low-level jet stream, I've annotated its axis on the 14Z analysis of 850-mb wind barbs and isotachs below.

 **The 14Z analysis of winds and isotachs on July 14, 2010, showing the axis of a low-level jet stream.**

The 14Z analysis of 850-mb winds and 850-mb isotachs (color-filled, in knots) on July 14, 2010. Note the fastest winds over northwest Iowa and Nebraska, where wind speeds exceeded 50 knots. This closed isotach marked the core of a low-level jet stream over the Northern and Central Plains.

Credit: Penn State University

To get your bearings, wind barbs designate 850-mb winds and color-filled contours indicate 850-mb isotachs (in knots). I've annotated the axis of the low-level jet stream, which passes through the core of its fastest winds, with

a white arrow. In this particular case, the core of the low-level jet stream was marked by the area where speeds exceeded 50 knots in western Iowa and eastern Nebraska.

This particular low-level jet stream had its [roots over the Gulf of Mexico](#) so there's no doubt that the low-level jet stream was carrying a rich supply of moist air. With such high wind speeds associated with the moist, low-level jet stream, it stands to reason that there would be a spike in moisture convergence in the lower troposphere. We can confirm that with the 14Z analysis of deep moisture convergence from SPC (below), which shows a pocket of deep moisture convergence (red contours) coincident with the ongoing thunderstorms. Revisiting the 14Z 850-mb isotachs above, note that there is some speed convergence associated with the low-level jet stream (fast winds transition to slower winds over Minnesota). Moreover, wind barbs over Minnesota are also confluent, and, in this case, the confluence adds to the overall pattern of convergence.

14Z analysis of deep moisture convergence on July 14, 2010.

The 14Z analysis of deep moisture convergence on July 14, 2010. The pocket of deep moisture convergence in northern Minnesota and northwest Wisconsin, marked by solid red contours, corresponded to an ongoing cluster of thunderstorms. The thin green contours indicate the average mixing ratio in the lowest 100 mb of the troposphere.

Credit: Storm Prediction Center

Note that I opted to look at deep moisture convergence instead of surface moisture convergence because SPC's deep moisture convergence product averages the moisture convergence over the lowest two kilometers of the troposphere. Thus, this field often extends high enough to capture the impact of the low-level jet stream (keep in mind that the standard height for 850 mb is 1.5 kilometers). In other words, in cases where there's elevated convection, looking at analyses of surface moist convergence is probably not the way to go, because unstable parcels are feeding into a storm's updraft [along a sloping frontal surface](#), usually a few thousand feet from the earth's surface.

The bottom line is that with moisture convergence occurring near 850 mb, and with upper-level divergence associated with vorticity maxima to the east of the 500-mb closed low that we noted earlier, the stage was set for deep, moist convection (in this case, elevated thunderstorms) over northern Minnesota and northwest Wisconsin on July 14, 2010.

This case gives you a taste of how elevated convection can develop in concert with a low-level jet stream and 850-mb warm advection. But, low-level jet streams can have another important consequence even outside of elevated convection situations. Let's explore.

Low-level Jet Streams and Wind Shear

Speedy winds associated with low-level jet streams can dramatically boost vertical wind shear in the lower troposphere. Because friction slows the wind at the surface of the Earth, fast winds at 850-mb are a good sign that there's a significant change in wind speed with increasing height in the lower troposphere. As we'll explore later, strong vertical wind shear in the lower troposphere, especially in the lowest kilometer, favors tornadogenesis whenever the storm environment supports the initiation of surface-based supercells.

To see an example of a low-level jet stream that caught the eye of forecasters, February 24, 2016 provides a great example. The 20Z Rapid Refresh analysis of 850-mb heights, winds, temperatures, and dew points (below) reveals a robust low-level jet stream along the Eastern Seaboard, to the east of a strong surface low-pressure system centered over the Ohio (check out the [21Z surface analysis](#)). Note the strong height gradient driving wind speeds near 70 knots as far north as Pennsylvania. Clearly, this low-level jet stream was ushering moist air northward, as evidenced by the northward bulge in 850-mb dew points greater than 10 degrees Celsius (shaded in green to indicate moist air).

20Z analysis of 850-mb heights, winds, temperatures, and dew points from February 24, 2016.

The 20Z 850-mb analysis on February 24, 2016. Dark contours denote 850-mb heights. Dashed blue and red contours are 850-mb isotherms. Wind barbs represent 850-mb winds. Contours filled with green shading indicate areas with dew points greater than 10 degrees Celsius. Note the potent low-level jet stream along the Eastern Seaboard (winds near 70 knots), and the northward surge of higher dew points that coincided with it.
Credit: Storm Prediction Center

Low-level jet streams of this magnitude (winds near 70 knots) are almost unheard of in the Middle Atlantic States in February. Noting this, forecasters realized that wind shear in the lower troposphere would be quite strong. The [20Z analysis of vertical shear between the surface and an altitude of one kilometer](#) showed that there was a whopping 30 to 60 knots of shear in that layer in the Middle Atlantic States. Recognizing the risk of rotating updrafts in thunderstorms (a component in the development of tornadoes), SPC wisely had [issued tornado watches from the Carolinas to Pennsylvania and New Jersey](#), and indeed, the [storm reports for February 24 included 27 tornado reports](#). Two of those tornadoes occurred in Pennsylvania, which marked only the second and third tornadoes on record in the state during February since 1950.

The bottom line is that you should make an analysis of the 850-mb pattern part of your forecasting routine so that you can keep an eye on temperature advection (particularly warm advection / overrunning), and to detect low-level jet streams (regions of strong height gradients and fast winds). Sometimes these two things go hand-in-hand, but either way the supply of lower tropospheric moisture and the increased low-level vertical wind shear in regions with low-level jet streams are important considerations for forecasters. On February 24, 2016, there's no doubt that the presence of the low-level jet stream helped alert forecasters to an out-of-season tornado risk in the Middle Atlantic States.

So, the 850-mb big-picture pattern should be a crucial part of your forecasting process when trying to predict thunderstorms -- either surface-based or elevated. While areas of 850-mb warm advection / overrunning can help identify hot spots for elevated convection to develop, 850-mb warm advection alone won't make for elevated thunderstorms. In the next section, we're going to look at all of the ingredients needed for elevated convection. Read on.

[< Prefrontal Troughs and Confluence Elevated Convection >](#)



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- [About](#)
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- [People](#)
- [Resources](#)
- [Services](#)
- [Login](#)

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